

Name: _____ Date: _____

Period: _____

Spiral Review

1. Let $f(t) = 3t^2 - 12t$.

(a) Is f increasing or decreasing on the interval $[0, 2]$? How do you know? (Hint: compare $f(0)$ and $f(2)$.)

(b) Is f increasing or decreasing on the interval $[2, 5]$?

2. Compute the average rate of change of $g(t) = t^2 - 4$ on the interval $[1, 4]$. Show your work.

3. Two particles both visit the point $(3, 5)$ in the plane. Particle A arrives when $t = 1$; Particle B arrives when $t = 2$. Explain why this is possible for parametric functions but is *not* possible for a function $y = f(x)$ (where $x = 3$ would force a unique y -value).
